

MD. ARIF SHEKH

Computer Science & Engineering | AI / Machine Learning & Full-Stack Developer

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Final-year Computer Science & Engineering student at IUBAT (CGPA 3.69/4.00) specializing in **machine learning**, **computer vision**, and **full-stack & backend development**. Two-time **1st Runner-Up as team lead** at national AI competitions — including the **bKash-presented NSUCEC Cybernauts Datathon 2026** — and **first author** of a submitted conference paper and an in-preparation thesis journal paper. A confident presenter and team leader seeking a **challenging role** where I can apply strong problem-solving, communication, and engineering skills to build impactful software and AI-driven solutions.

PROJECTS

Customer Churn & Survival Prediction — bKash NSUCEC Cybernauts Datathon1st Runner-Up (Team Lead) · 2026

Python · Polars · LightGBM / XGBoost / CatBoost · Optuna · SHAP · scikit-survival

- Processed **200M+ transactions out-of-core** with Polars (lazy scan) and engineered **152 behavioral features**; trained a 5-fold LightGBM/XGBoost/CatBoost ensemble reaching **ROC-AUC 0.985**.
- Extended to a time-to-event survival model predicting *when* customers churn (isotonic calibration, monotone survival) — **c-index 0.765, Integrated Brier Score 0.0132, 3.3x top-decile lift**; led the team to 1st Runner-Up at the national grand finale.

RoadVision — Bangladeshi Highway Vehicle Detection (DUET CSE Carnival AI Hackathon)1st Runner-Up (Team Lead) · 2026

Python · YOLO11 (l + x) · YOLOv12-X / RF-DETR · Weighted Box Fusion · TTA · Ultralytics

- Detected & classified **13 native vehicle classes** on Bangladeshi highway CCTV (severe class imbalance, small objects); fine-tuned dual YOLO11l + YOLO11x at 1280px with mosaic/mixup/copy-paste augmentation.
- Fused detectors via **Weighted Box Fusion** + multi-scale TTA — private-leaderboard **mAP@0.5 0.656** (val 0.867), ranked #2 nationally, leading Team “Straw Hat”.

CoWork — Coworking-Space Booking REST API (IUT ICT Fest Hackathon)Team Straw Hat · 2026

Python 3.11 · FastAPI · SQLAlchemy · JWT (access + refresh) · Pydantic · Docker · pytest

- Built a **multi-tenant REST API** for room booking — **JWT authentication** with access/refresh-token rotation, role-based access (admin / member), dynamic pricing, a tiered refund policy, and paginated, filterable queries.
- Hardened the service for **concurrency and tenant isolation** while preserving strict API-contract compliance; **45 behavioral tests passing** on both local and **Docker** deployments.

Pedestrian Crossing-Intention Prediction — Undergraduate ThesisFirst-Author Journal · 2026

Python · PyTorch · two-stream BiLSTM · YOLO26 · ByteTrack · PIE dataset

- Built a two-stream (bounding-box + ego-speed) BiLSTM on the PIE dataset under a **leakage-free protocol** — **ROC-AUC 0.932, accuracy 0.883 at 0.575 ms/window**, matching far heavier multimodal models.
- Shipped a real-time demo (YOLO26 → ByteTrack → BiLSTM intent overlay); **first-author** manuscript targeted for **journal publication in Aug–Sep 2026**.

MedKAN — Brain-Tumor MRI ClassificationFirst-Author Paper · 2026

Python · PyTorch · Kolmogorov–Arnold Networks · timm · Vision Transformers

- Proposed **MedKAN-B**, a Kolmogorov–Arnold Network architecture for 4-class brain-tumor MRI classification, benchmarked against CNN and Vision Transformer baselines with emphasis on **low-data regimes** (100 / 50 / 25% training fractions); **first-author** paper submitted to **SPICSCON 2026**.

Bengali LLM Hallucination Detection — IUT ICT Fest DatathonOngoing · 2026

Python · NLP · Transformers · Kaggle

- Building an **offline-reproducible Bengali hallucination detector** (binary faithfulness classification of prompt–response pairs) for a 20-day IUT datathon — constrained to run fully inside Kaggle with no hosted LLM APIs.

Aamal Foods — Full-Stack E-Commerce Platform2026

Django · Python · Bootstrap · JavaScript · Stripe & SSLCommerz

- Built an end-to-end storefront — product catalog across 7 categories, cart, wishlist, order tracking, and authenticated user profiles.
- Developed a custom admin dashboard of **50+ endpoints** (sales analytics, inventory & expiry tracking, delivery & invoicing, staff management) with **dual payment gateways** (Stripe, SSLCommerz) integrated.

Dry Run — Browser-Based Robotic-Arm Control Suite (Techathon 2026, IUT Robotics Society)Finalist · Team Straw Hat · 2026

TypeScript (strict) · React · Zod · Zustand · three.js · Groq LLM · Vitest

- Engineered a safety-critical control system on a **single-gate command pipeline** — dashboard, keyboard, joystick, voice and an LLM planner all routed through one typed **validate()** function — with **62 unit tests** and a clean, documented architecture demonstrating **SOLID and design-pattern** discipline.

AI-Based Loan-Approval Expert System — hybrid rule-based + Bayesian + Random Forest pipeline on a 4,269-applicant dataset (CSE 4383 – Artificial Intelligence).

Python · scikit-learn

Library Management System — full-stack app with authentication, role-based access, book/loan management, and admin dashboard. C# · ASP.NET Core MVC

OSPF Dynamic-Routing Migration — designed & configured single-area OSPF across 3 sites (Dhaka, Chattogram, Sylhet); RIP-vs-OSPF evaluation. Cisco Packet Tracer

Daily Minimum Temperature Forecasting — time-series forecasting on real climate data (Data Science course). Python · pandas · Jupyter

EDUCATIONAL QUALIFICATIONS

B.Sc. in Computer Science & Engineering — Expected 2026 | IUBAT (International University of Business Agriculture and Technology) | CGPA: 3.69/4.00

Higher Secondary Certificate (Science) | Rajbari Government College | GPA: 5.00/5.00

Secondary School Certificate (Science) | Rajbari Government High School | GPA: 5.00/5.00

CORE COMPETENCIES

- Machine Learning & Deep Learning
- Full-Stack Web Development
- Data Analysis & Feature Engineering
- Computer Vision
- REST API & System Design
- Model Evaluation & Tuning

TECHNICAL SKILLS

Languages: Python · C / C++ · JavaScript / TypeScript · C# · SQL

ML & Data: PyTorch · scikit-learn · LightGBM · XGBoost · CatBoost · Optuna · SHAP · Polars · pandas · OpenCV · YOLO11/26 · ByteTrack · Vision Transformers · KAN · NLP · survival analysis

Web & Backend: Django · Django REST · FastAPI · SQLAlchemy · Next.js · React · TailwindCSS · Bootstrap · REST APIs · WebSockets

Databases & Tools: MySQL · PostgreSQL · SQLite · Docker · Git & GitHub · Linux/Unix · Kaggle · Jupyter · VS Code · Cisco Packet Tracer

SOFT SKILLS & LANGUAGES

Soft Skills: Team Leadership · Presentation & Public Speaking · English Communication · Project & Time Management · Teamwork & Collaboration · Problem Solving · Adaptability

Languages: Bangla — Native · English — Professional (fluent in speaking, writing & presentation; Grameenphone Academy “English for Daily Use” — Certificate of Excellence)

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

1st Runner-Up (Team Lead) | bKash NSUCEC Cybernauts Datathon 2026 — a national machine-learning datathon on customer-churn prediction.

1st Runner-Up (Team Lead) | DUET CSE Carnival 2026 AI Hackathon — vehicle detection on Bangladeshi highway CCTV.

Finalist (Team Straw Hat) | Techathon 2026, IUT Robotics Society — browser-based robotic-arm control suite.

Participant (Team Straw Hat) | IUT ICT Fest 2026 Hackathon & Datathon — backend API engineering and Bengali LLM hallucination detection.

Participant | The Infinity AI BuildFest 2026 — CloudCamp Bangladesh × BRAC University.

Participant | Idea Competition, IEEE ICADHI 2026 — Daffodil International University.

Attendee | IEEE SSIT Seminar “AI & Robotics for Sustainable Innovation” — BUET, Dhaka.

Certificates | Full archive of competition, achievement & participation certificates: drive.google.com/.../certificates.

REFERENCES

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